

Rhythm × Topology: which key fits which lock

	Cyclic graph (Sudoku, n-queens, graph coloring)	Tree graph (arithmetic DAGs, dependency graphs)
Rotational rhythm (π -cycled RoPE)	v98 ✓ 79% puzzle accuracy Exponential energy decay (rate ≈ 0.18 nats/breath)	v99 ✗ 9% (chance floor) Flat CE ladder No depth correlation
Topological rhythm (staged masks)	Future work (Prediction: fails — cyclic graphs have no natural DAG order)	v100/v101/v103 ✓ 40–48% cell accuracy +18 pt K=1→K=10 gap Depth-correlated

Success (diagonal)

Failure (off-diagonal)

Untested prediction